BTS-I-11-19-1447 Reg.No. [redacted]

A

B.Tech. Degree I Semester Examination November 2019

CE/EE/ME/SE 19-200-0102 A ENGINEERING CHEMISTRY

(*2019 Scheme*)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(8 \times 3 = 24)$

I. (a) Discuss the characters of an acceptable wave function.
(b) Using Schrodinger wave equation derive the energy of a particle in 1-D box.
(c) Explain the phenomenon of fluorescence.
(d) Write the principle behind IR spectroscopy.
(e) Give any three statements of second law of thermodynamics.
(f) Describe the phase diagram of simple eutectic system.
(g) Explain the mechanism of polymerization when organic peroxides are used as catalyst.
(h) Give the preparation, properties and uses of (i) PVC (ii) PET.

PART B

$(4 \times 12 = 48)$

II. (a) Derive Schrodinger equation from classical wave equation. (6)
(b) State and explain the Huckel's rule of aromaticity. (6)

OR

III. (a) Give the hydrogen atom wave functions and their solutions. (6)
(b) Explain with suitable plots of the Pi-molecular orbital wave functions and their energies of butadiene. (6)

IV. (a) Explain the principle and the chemical shift pattern in NMR spectroscopy. (6)
(b) Discuss the possible electronic transitions, their regions of absorption and the shifts in positions of absorption in electronic spectroscopy. (6)

OR

V. (a) Derive energy and explain the nature of the spectrum of a rigid rotor. (6)
(b) Describe spin-spin splitting patterns in NMR spectrum of ethyl alcohol and dimethyl ether. (6)

VI. (a) Derive a relation connecting free energy change with equilibrium constant of a reaction. (6)
(b) Explain the phase diagram of one component water system. (6)

OR

VII. (a) State and establish first law of thermodynamics with a suitable derivation. (6)
(b) Derive Kirchhoff's equations at constant pressure and volume. (6)

VIII. (a) Give the co-ordination and cationic mechanism of polymerization. (6)
(b) Explain the preparation, properties and uses of silicone polymers. (6)

OR

IX. (a) Account on manufacture, setting and hardening of cement. (6)
(b) What are refractories? Explain the different classes of refractories with examples. (6)

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